pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Z-Scores and Percentile Ranks

WEEK 2 · DAY 3

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From moves to levels

Day 7 measured how much a series moves. Today asks where its level sits, compared with its own last year.

z-score standard deviations from the window's mean

percentile rank share of the window at or below today

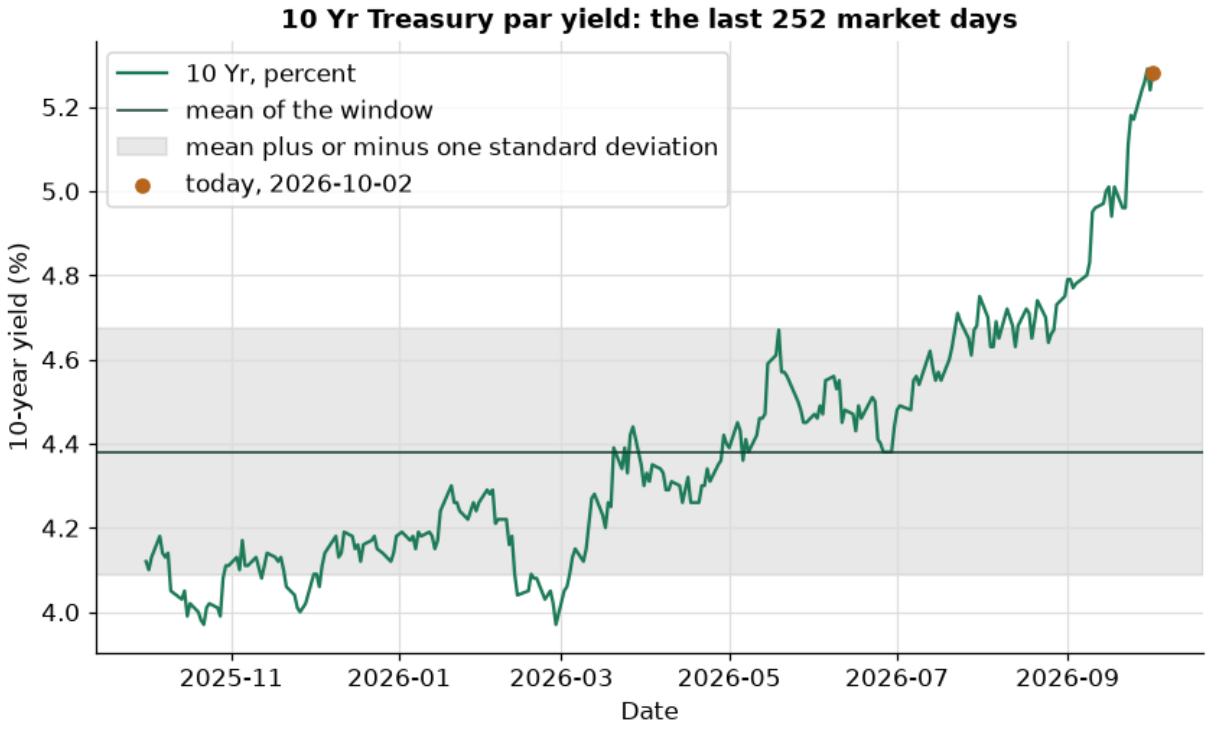
Both say where today sits in its own history. Never a signal, never cheap or rich, never due to revert, and no threshold.

No live cell today: every number comes from the Treasury file.

Week 2: time series measures

Day 6	Changes in bp and percent
Day 7	Rolling windows, volatility
Day 8	**Z-scores, percentile ranks**
Day 9	Slopes and butterflies
Day 10	Case study: three years

Where does today sit in its last 252 days?



Low 3.97, high 5.29, today 5.28 percent

90.0 bp

above the window's mean

3.0752

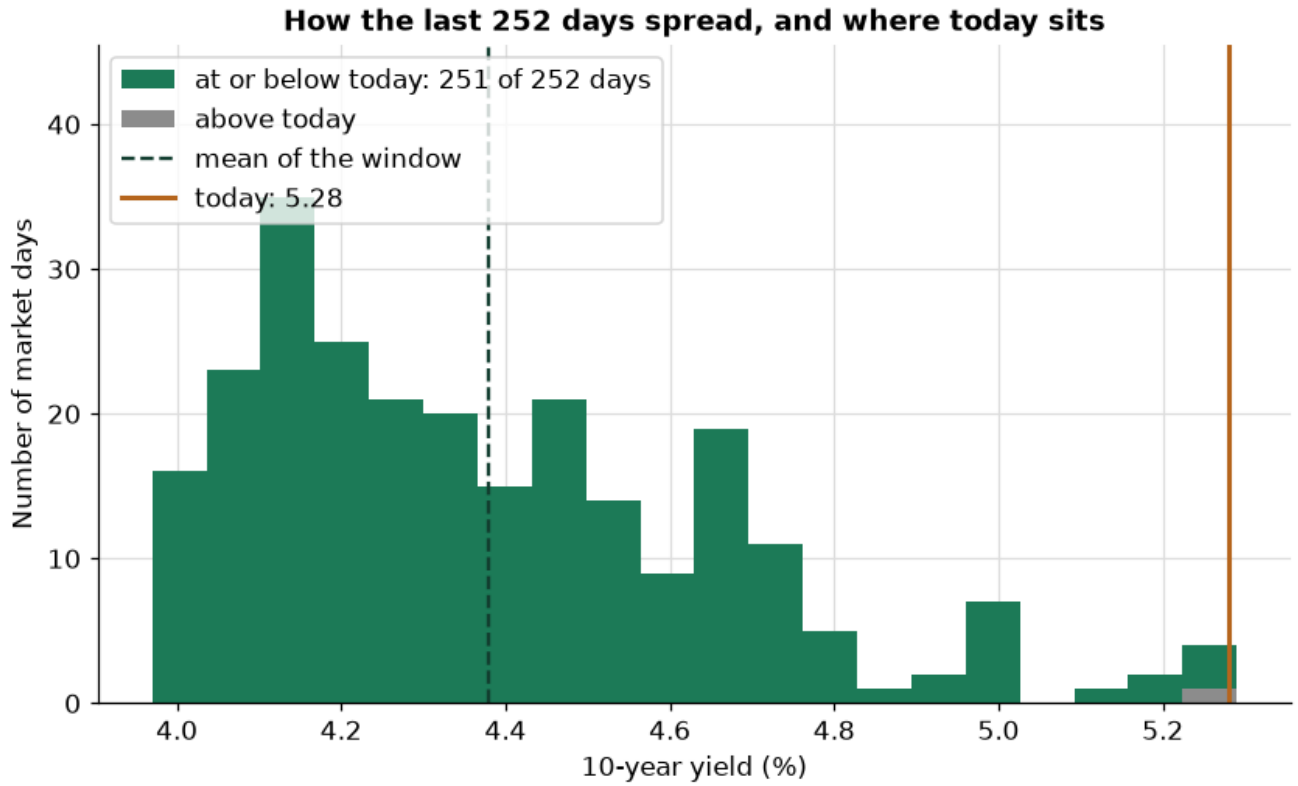
standard deviations: 90.0 bp over 29.3 bp

99.6 percent

251 of 252 days at or below today

Three descriptions of the past 252 rows. None says anything about tomorrow.

The window's values, and where today sits



Reading the bars

Each bar counts market days in that range. The green bars are the 251 of 252 days at or below today: the percentile rank's numerator.

The dashed line is the mean the z-score measures from.

The z-score and its window

```
z = (x_today - window mean) / window sample std
# window: trailing 252 rows, today included
# ddof=1 (STDEV.S); flat window: NaN
```

No unit: a yield in percent and a slope in bp both give plain numbers. Not a probability: the course makes no assumption about the shape of the distribution.

Is today in its own window?

Today included (this course)	**3.0752**
The 252 rows before today	**3.1470**

Same data, same length, a different number.

Neither choice is wrong. A number with no definition beside it is.

Today in its window, or left out



Two definitions

On the last day: 3.0752 with today in, 3.1470 with today left out. The lines follow each other and never match.

Neither is wrong. Write the definition beside the number.

The percentile rank and ties

```
rank = 100 x (window values <= today) / count
# today counts itself; ties count as at or below
```

Ten values: 5, 7, 7, 9, 11, 12, 3, 7, 10, 8. At or below 7: the three 7s, the 5, and the 3. Five of ten, 50 percent.

Yields are quoted to two decimals, so ties are common in every window.

Method for the three 7s	pandas	Excel (tested)
Lowest tied rank	min: 3	RANK.EQ: 3
Average tied rank	average: 4	RANK.AVG: 4
Count at or below	max: 5	COUNTIF

Only method="max" counts at or below: rank(pct=True, method="max").

The floating point trap

```
(4.12 - 3.80) * 100   ->  32.00000000000003
(4.13 - 3.81) * 100   ->  31.999999999999986
# two 32 bp slopes, not equal in binary
```

Excel's COUNTIF compares at 15 significant digits: at or below the second slope, it counts both, 100 percent (tested).

Worst day, 2s10s at 52 bp: 5 window values equal exactly, 20 equal at 15 digits.

0 of 190

10 Yr rows where plain pandas differs from COUNTIF

74 of 190

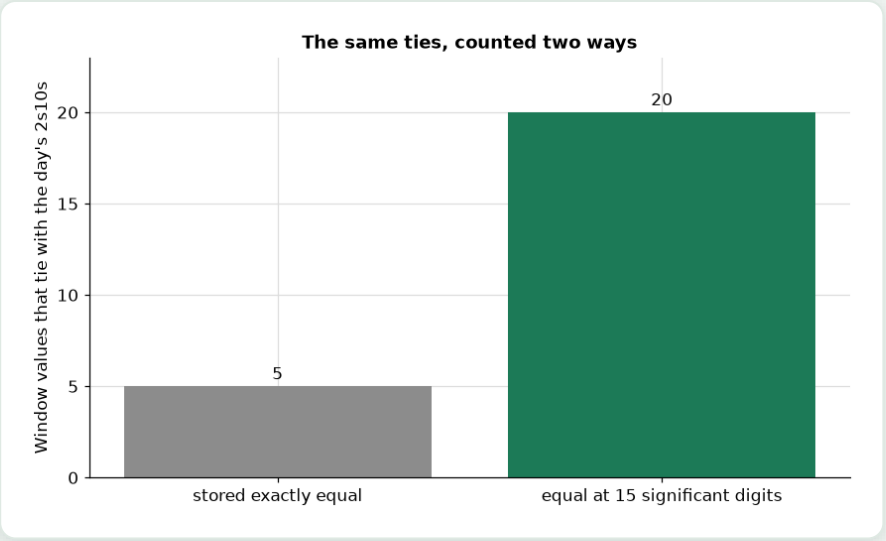
2s10s rows, by up to 5.95 points

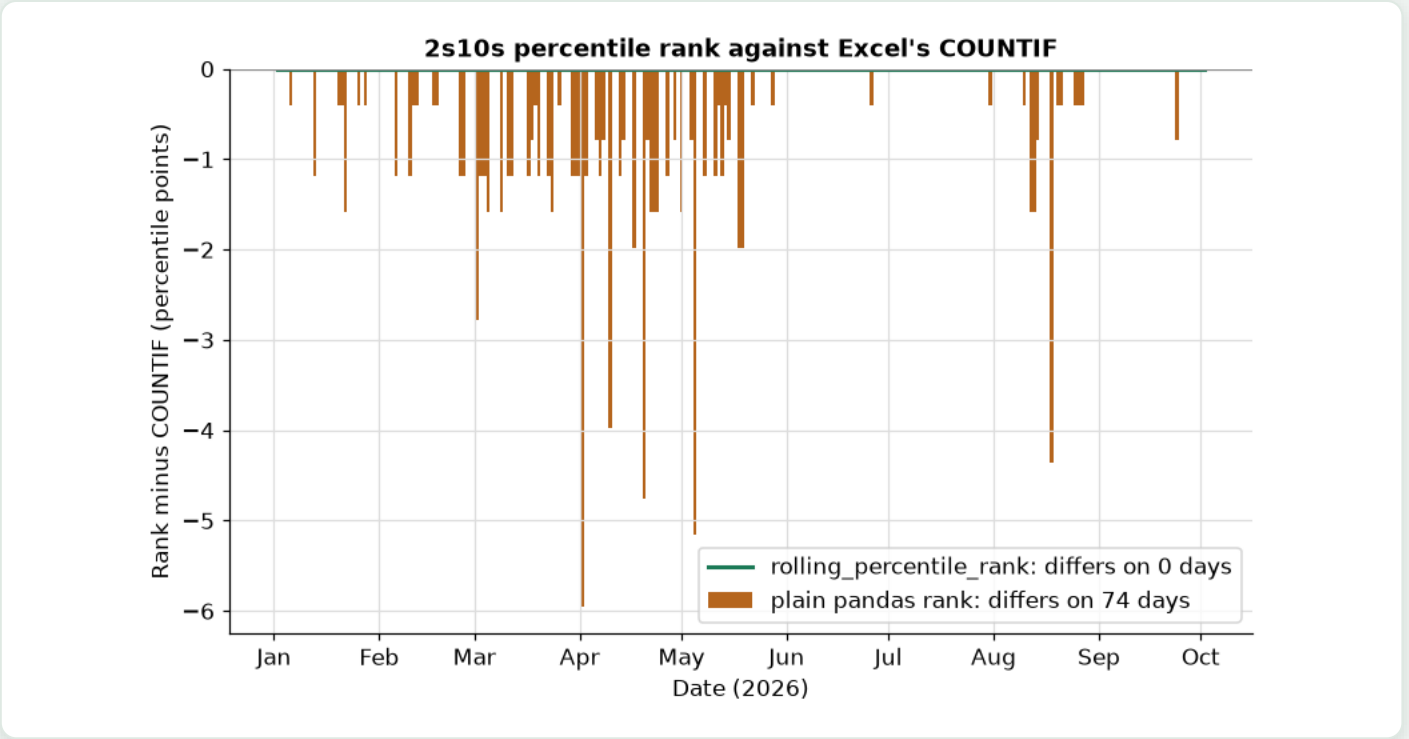
0 of 190

2s10s rows after rounding to 15 significant digits first

Any equality on computed numbers (==, a rank, a merge, COUNTIF) can split a tie.

The tie split, drawn





Left: 20 ties at 15 digits, 5 exactly. Right: plain pandas misses Excel's ties on 74 days; the rounded rank on 0.

PERCENTRANK.INC is a different measure

On the ten values	for 7	for 8
Share at or below (COUNTIF/COUNT)	50 percent	60 percent
PERCENTRANK.INC (tested)	0.222	0.555
PERCENTRANK.EXC (tested)	0.272	

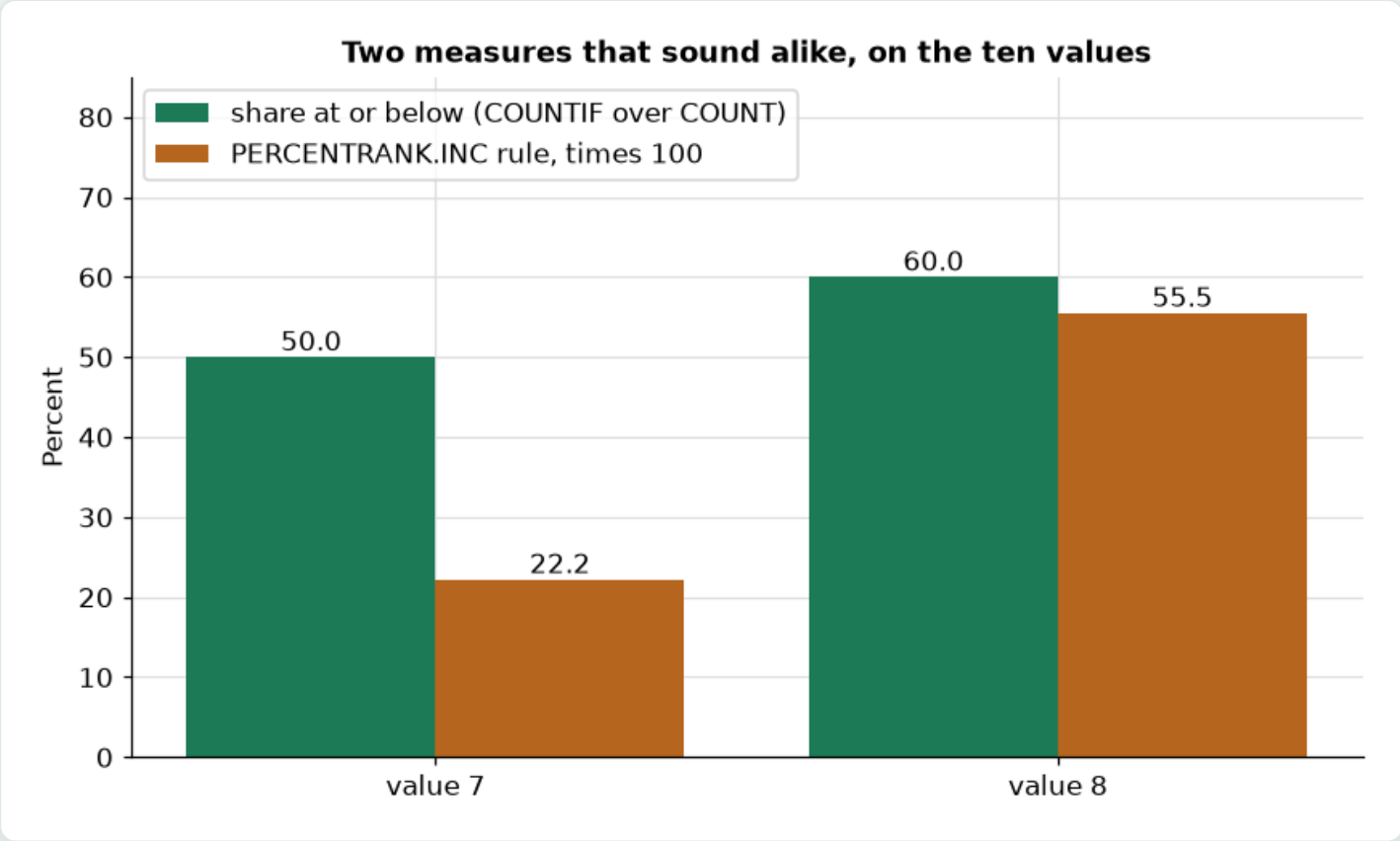
What PERCENTRANK.INC computes

Values strictly below, over n - 1, cut (not rounded) to three decimals. For 7: two of nine, 0.222.

On the reference rows the rule equals Excel's column on 190 of 190 rows, and still differs from the percentile rank by up to 5.02 points (10 Yr) and 10.13 (2s10s).

Correct for its own question. The share at or below is COUNTIF over COUNT.

Two measures that sound alike, drawn

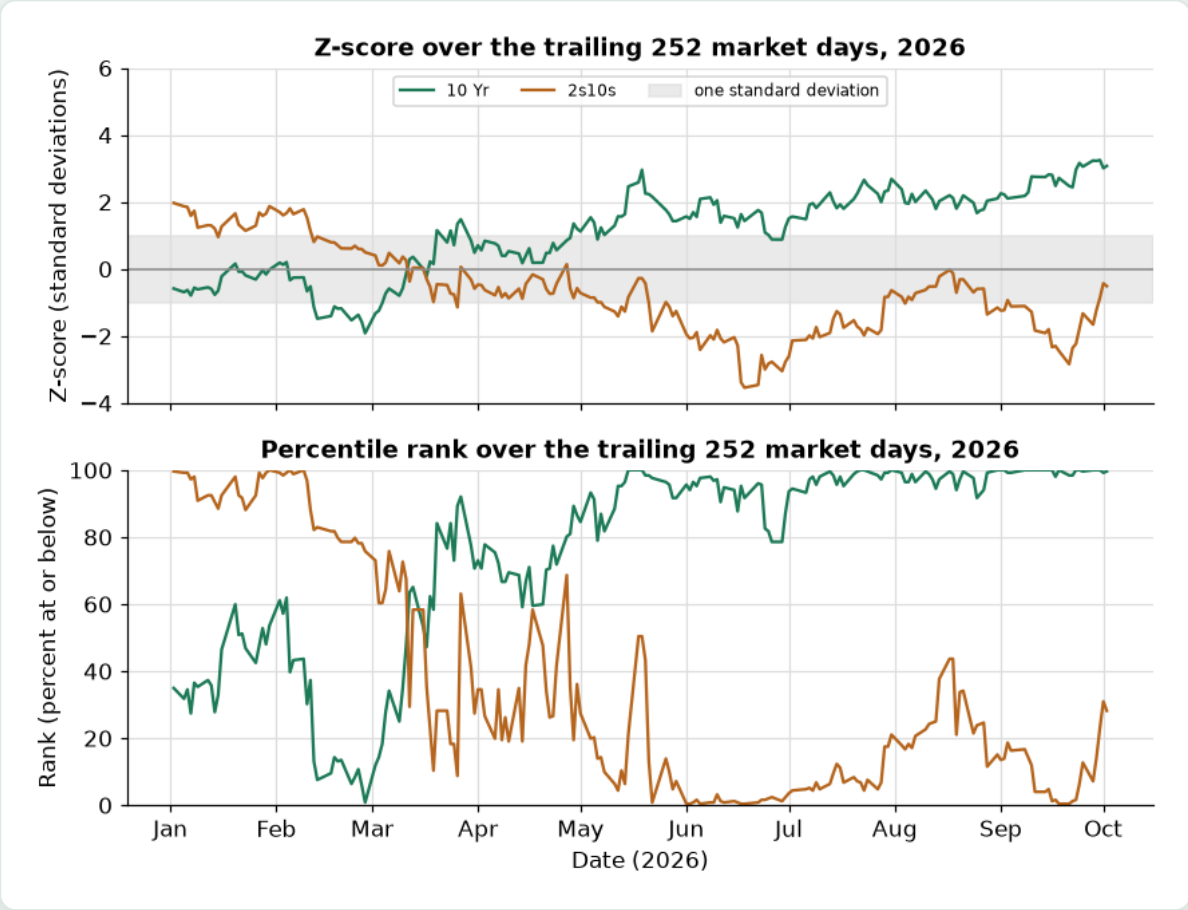


Same data, two questions

For 7: 50 at or below, against 22.2 for the rule. The three 7s count in one and not in the other.

Use COUNTIF over COUNT for the share at or below.

Both measures on the 10 Yr and 2s10s, 2026



On 2026-10-02

10 Yr: 5.28 percent, 3.08 standard deviations above the mean of its last 252 market days; 99.6 percent of those days at or below it.

2s10s: 45.0 bp, 0.51 standard deviations below its mean; 28.2 percent of the window at or below it.

Each sentence says where a number sits. Neither says where it goes next.

No lookahead: cut at a date, the answers up to it do not change. 35 tests pass.

Words that describe, words that forecast

Describe (this course)	Forecast or advise (never)
3.08 sd above the mean of 252 rows	stretched, extreme
99.6 percent of the window at or below	overbought, oversold
the low and high of the window	cheap, rich, attractive
up so many bp over the month	due to revert, will fall
the window is 252 rows, today included	a signal, a buy, a sell

No threshold: no line at a z-score of 2 or the 90th percentile. A threshold is a decision rule.

The window's length is a choice, today's place in it is a choice, and the past year may be nothing like the next.

Excel side by side

Excel side by side (tested in Excel)

`=STANDARDIZE(C254,AVERAGE(C3:C254),STDEV.S(C3:C254))`	3.075234
`=STANDARDIZE(C254,AVERAGE(C2:C253),STDEV.S(C2:C253))`	3.147044 (today left out)
`=COUNTIF(C3:C254,"<="&C254)/COUNT(C3:C254)*100`	99.603175
`=PERCENTRANK.INC(C3:C254,C254)`	0.996 (a lookalike)
`=STANDARDIZE(A3,AVERAGE(A1:A3),STDEV.S(A1:A3))`	#NUM! (5, 5, 5)

The sheet: dates in A, 2 Yr in B, 10 Yr in C, rows 2 to 254; 2s10s in D as =(C2-B2)*100. Write the formula in the first full window's row and fill down: each row gets its own trailing window.

Tested in Excel 16.0 build 20527, 2026-11-02. Python equals every value to within 1e-12.

Save the day: git grep

```
git grep -n "rolling(" -- marketdata.py
413: vol = changes_bp.rolling(window, min_periods=window).std(ddof=1)
434: mean = series.rolling(window, min_periods=window).mean()
435: std = series.rolling(window, min_periods=window).std(ddof=1)
460: return rounded.rolling(window, min_periods=window).rank(pct=True, ...
```

```
git grep -n "window: int = " -- marketdata.py
398:def rolling_vol_bp(changes_bp: pd.Series, window: int = 63, annualise: ...
420:def rolling_zscore(series: pd.Series, window: int = 252) -> pd.Series:
441:def rolling_percentile_rank(series: pd.Series, window: int = 252) -> ...
```

Before you change a default, find every place it lives. git grep searches only what Git tracks.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the z-score uses the whole series' mean and standard deviation: 2026-06-30 reads 1.4157 in full, 1.4886 cut there.

Code that runs is not the same as code that is right.

What you can do now

- Say where today sits three ways: bp from the mean, z-score, percentile rank
- Z-score over a trailing 252-row window that includes today
- Percentile rank: share at or below, ties counted, at 15 significant digits
- PERCENTRANK.INC is a lookalike; COUNTIF over COUNT is the share
- Prove no lookahead; describe a level, never forecast it

Week 2, Day 4 Slopes and butterflies: signs, functions, and the rolling z-score of each